MOHAMMED SHABAZ S

Embedded Engineer | Automotive Embedded Systems

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SUMMARY

Final-year ECE student (CGPA 8.38) focused on embedded and automotive firmware using ARM Cortex-M microcontrollers, finite-state machines, and non-blocking Embedded C/C++ on ESP32 and STM32. Internship experience at Nokia and iHelp Robotics; completed LTE/5G self-study.

TECHNICAL SKILLS

Programming: C, C++, Python, SQL (Intermediate); Java (currently learning)
Embedded: Embedded C, Microcontrollers (Intermediate); ARM Cortex-M4, FSM, non-blocking design, GPIO, I2C, ADC, timers, sensors, debugging
Mobile Networks: LTE/5G architecture, EPC/RAN, interfaces, call flows, QoS/RRM, EN-DC, mobility, VoNR, O-RAN, troubleshooting, test automation (completed self-study)
Tools & Foundations: Git/GitHub; TensorFlow Lite; DSA, DBMS, OS, Computer Networks (Basic)

EXPERIENCE

Student Intern 16 September 2026 – Present
Nokia Solutions and Networks India *Bangalore*

- Working as a Student Intern at Nokia.

AI Research Intern – Deep Learning & Model Development 23 March 2026 – 23 August 2026
iHelp Robotics Private Limited · Project: MITRA *Remote*

- Improved live camera-stream reliability in the MITRA assistive Android app across 2 video sources (hardware WiFi/RTSP and phone-camera fallback): cut stream search delay, persisted the last working transport, and added refresh/stall recovery.
- Engineered a visual-freeze watchdog plus 3 status/guard mechanisms — stream-status and cloud-status panels and a WebSocket-gated send guard — and fixed a live-refresh loop that recurred after 5 minutes.
- Hardened offline voice commands (offline-preferred recognition with model recovery and microphone retry backoff) and verified builds across multiple devices with Gradle test/lint/assemble; delivered 6+ app-side improvements while model and backend stayed with a separate team.

Student Intern 1–30 June 2026
IEEE EMBS Pune Section *Remote*

PROJECTS

Automotive Body Control Module GitHub
ESP32, Arduino/C++, Finite-State Machine, GPIO, I2C OLED

- Programmed an ESP32 body-control module with a 3-state (OFF/ACC/ON) ignition FSM driving 6+ vehicle functions: turn indicators, synchronized hazard, brake logic, buzzer feedback and an OLED dashboard.
- Architected a non-blocking millis()-based scheduler (zero delay() in the main loop) with 30 ms brake debouncing, throttled 10 Hz I2C OLED refresh, a clean GPIO abstraction and edge-triggered serial logging for deterministic real-time behavior.

Smart Wellness Desk Assistant GitHub
STM32 Nucleo-L476RG (ARM Cortex-M4), STM32 HAL, STM32CubeIDE, Embedded C · College team project

- Interfaced ultrasonic presence and 12-bit ADC temperature sensors, an I2C SSD1306 OLED and timer-driven buzzer alerts on an STM32 Nucleo-L476RG (ARM Cortex-M4) using STM32 HAL, coordinated by a non-blocking finite-state machine; wired, tested and debugged in STM32CubeIDE as part of a college team.

VisionPay — Real-Time Indian Currency Detection GitHub
Python, OpenCV, TensorFlow Lite, MobileNetV2, Text-to-Speech

- Created an offline, real-time tool that recognizes Indian currency from a live webcam and speaks the result aloud for visually impaired users.
- Collected ~400 images per denomination across 6 classes (Rs.10–Rs.500) and trained a MobileNetV2 classifier converted to TensorFlow Lite, reporting ~93% validation accuracy.
- Strengthened real-time reliability with 4 mechanisms: confidence/margin gating, temporal smoothing, a background class, and an auto-count mode.

EDUCATION

B.E. Electronics & Communication Engineering Expected 2027
Ballari Institute of Technology and Management (BITM), Ballari *Final Year · CGPA 8.38*

LEADERSHIP & ACTIVITIES

- Vice-Chair, IEEE CAS Society, BITM; Treasurer, BITM Robotics Club; Google Student Ambassador (2025).
- IEEE SPACE 2026 participant; STMicroelectronics Embedded AI workshop; Techzone Nationals 2K25 participant.

CERTIFICATIONS

Embedded Systems — Internshala Trainings (2025) · Programming in C and C++ with AI · SQL for Data Analytics with AI · Google Cloud Generative AI — SmartBridge/SmartInternz (2025) · Python Programming — EISystems (2024)